

Executive Summary

Highway construction projects overseen by the U.S. Department of Transportation's Federal Highway Administration (FHWA) often require the usage of concrete pavement. Compressive strength of concrete pavement is determined by a non-trivial combination of factors, including ratio of cementitious paste components, aggregates, and curing time. A linear regression model of concrete compressive strength is devised using the UCI Concrete Compressive Strength dataset of 1030 unique concrete instances. The top line model performance yielded an $R^2 = 0.624$ indicating that the model can explain 62.4% of the training data's variability. Analysis of model parameters finds that increasing cement content and curing age have the strongest positive impact on concrete compressive strength, while increasing water content has the strongest decrease on concrete compressive strength. The following recommendations for increasing compressive strength are given: use cement and slag in 1:1 ratio to increase compressive strength, use plasticizer over water to increase workability, and prefer aging over aggregates for optimizing strength.

Problem Context and Data Overview

Highway construction spans a large set of complex civil engineering challenges. In the U.S, the FHWA is responsible for overseeing these highway projects and provides technical literature and best practices. Pavement selection is one such design decision, affecting costs, road longevity, and safety of a highway. Concrete is one pavement well-suited for longevity and strength; however, the components ratios selected must be carefully balanced based on a highway project's specific goals. These goals can be separated roughly into two categories: organizational and engineering. Organizational goals primarily addressing logistical, real-world implications that result from design choices. Some examples of these include:

- Reducing the overall cost of materials used per mass or volume
- Reducing the total volume of materials required
- Reducing the number of required concrete component vendors
- Shortening the construction time for highway projects

Reducing the volume, mass, and variety of materials can proportionally save the FHWA organizational costs. Furthermore, logistical transportation costs are saved by having to haul less material and from fewer collection sites. Transportation also has indirect maintenance costs to damage to existing infrastructure from enduring heavy trucking loads. Longer curing time also increases construction project length, thereby increasing traffic and delays to drivers. Engineering goals address the project's technical scope. Examples of these include:

- Meeting strength requirements for safety compliance

- Extending the longevity of the concrete
- Minimizing the mass, volume, number and time required to achieve a compressive strength

Each concrete has a specific function. Cement and water are the primary materials in cementitious paste. When combined, the water hydrates the cement and gives it enough fluidity to be compacted and placed before setting. The total amount of concrete and water-to-concrete ratio affect early-stage strength and workability. Blast furnace slag and fly ash are both supplementary materials in the paste, and can help with late-stage strength development. Superplasticizer increases paste workability without adding water. Coarse and fine aggregates form the underlying skeleton of the concrete, determining the final volume and structural stability. Increasing the amount of aggregate can increase strength but also increase overall paste demand.

Solutions to optimizing both these organizational and engineering goals are highly variable as well as being sensitive to a given project's location and economics. Table 1 gives a statistical summary of concrete's component make-up for 900 unique samples. Components are measured in kilograms per cubic-meter, and have wide standard deviations. Curing age, in days, has a standard deviation of roughly 2 months. In some instances, components (blast furnace slag, fly ash, and superplasticizer) are not included.

Feature	Mean	Standard Deviation	Minimum	Maximum
Cement (kg per cubic-meter mixture)	278.16	106.16	102.0	540.0
Blast Furnace Slag(kg per cubic-meter mixture)	76.14	84.01	0.0	305.3
Fly Ash (kg per cubic-meter mixture)	57.52	64.23	0.0	200.1
Water (kg per cubic-meter mixture)	180.74	22.27	121.8	247.0
Superplasticizer (kg in a	6.87	5.98	0.0	32.2

Feature	Mean	Standard Deviation	Minimum	Maximum
cubic-meter mixture)				
Coarse Aggregate (kg in a cubic-meter mixture)	973.44	77.55	801.0	1145.0
Fine Aggregate (kg in a cubic-meter mixture)	772.18	81.87	594.0	992.6
Age (day)	46.04	61.28	1.0	365.0
Concrete compressive strength (MPa)	36.97	16.65	2.33	82.6

Table 1: Descriptive statistics of 900-sample training data

Figures 1 and 2 show the effects of varying cement mass and curing age on the concrete's compressive strength. Figure 1 shows a trend of increasing compressive strength with added cement. Note that this scatter plot does not hold all other variables fixed, which explains the compressive strength variance for a fixed mass of cement. Figure 2 shows that longer curing times are correlated with greater compressive strength. Concrete strength appears to develop quickly in early-stages while slowing down in the later-stages.

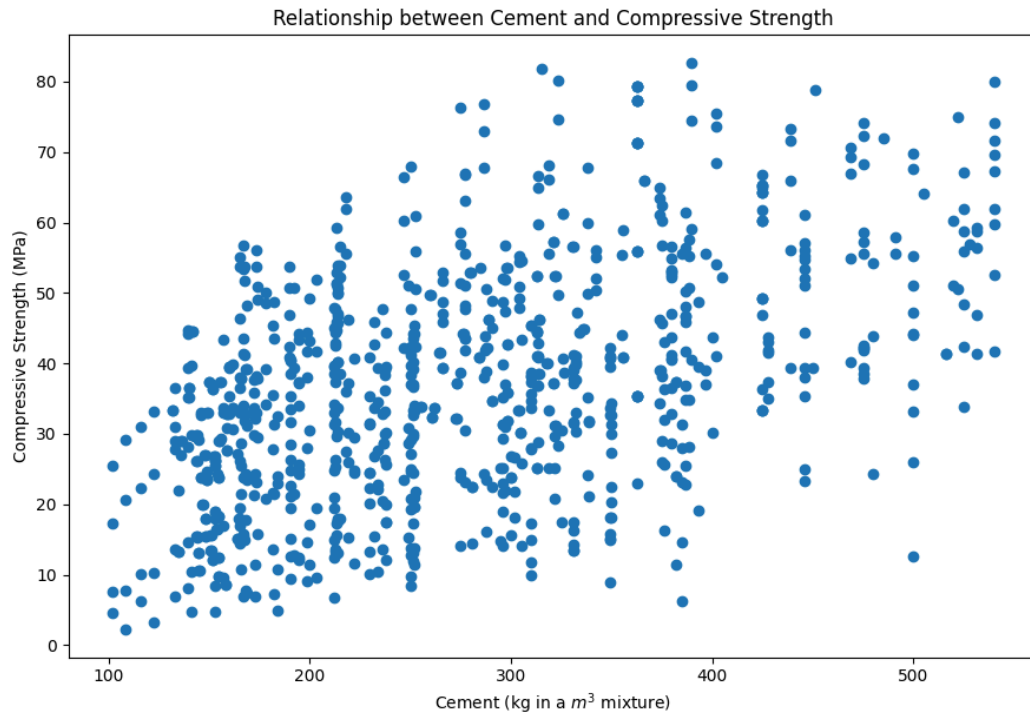


Figure 1: Relationship between cement and compressive strength

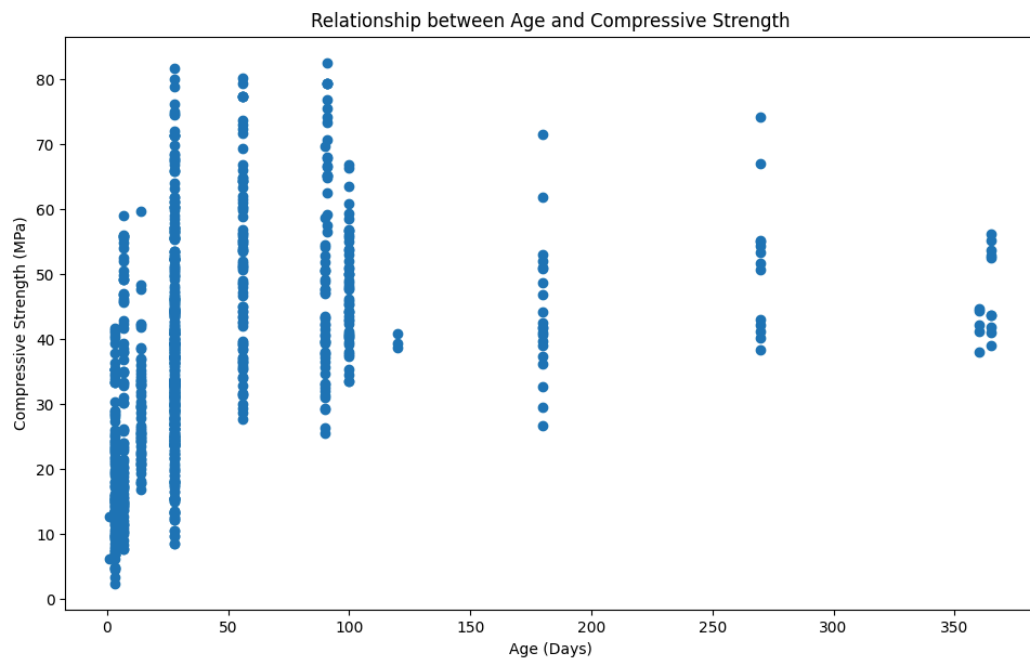


Figure 2: Relationship between curing age and compressive strength

Model Reliability and Analytical Findings

A linear model trained on 900 concrete samples is constructed to predict concrete compressive strength given masses of concrete components and curing time. Table 2 shows an estimated increase in compressive strength with a unit increase of each feature.

Feature	Increase in compressive strength per unit (MPa)
Cement (kg in a cubic-meter mixture)	0.1328453
Blast Furnace Slag(kg in a cubic-meter mixture)	0.12504038
Fly Ash (kg in a cubic-meter mixture)	0.10673935
Water (kg in a cubic-meter mixture)	-0.13313546
Superplasticizer (kg in a cubic-meter mixture)	0.11618198
Coarse Aggregate (kg in a cubic-meter mixture)	0.02863288
Fine Aggregate (kg in a cubic-meter mixture)	0.0340842
Age (day)	0.11876335

Table 2: Estimated increase in compressive strength for unit increase of feature

For each cement component, a unit increase is an addition of 1kg in a cubic-meter mixture. For curing age, a unit increase is an addition of 1 day. Each estimate assumes that all other features are held constant. For example, increasing cement by $1\text{kg}/\text{m}^3$ yields a 0.133 MPa increase in compressive strength, while increasing water by $1\text{kg}/\text{m}^3$ yields a 0.133 MPa decrease in compressive strength. For curing time, model coefficients mean that aging by 1 day yields a 0.112 MPa increase in compressive strength

The models are evaluated using the R^2 statistic, which assigns a percentage to the model's ability to explain the data's variation. Evaluating on the training data yields $R^2 = 0.624$ – i.e. the model can explain 62.4% of training data's variation. Evaluating on test data the model has never seen evaluates the model's predictive power. Using 130 test samples yields $R^2 = 0.351$. A model explaining 35.1% of variation means that the model does not generalize well to unseen data. A better model should be devised if predictive performance is required.

Analyzing the data also reveals trends of the concrete's physical properties. Figure 3 shows concrete compressive strength as a function of the water-to-cement ratio. This plot shows a visible negative correlation between the ratio and compressive strength. Also shown is a preference to use equal or more concrete than water. This is corroborated by the model, which

predicts that adding 1 kg/m^3 of cement or removing an equivalent amount of water both yield a 0.133 MPa increase in compressive strength. The material tradeoff for strength then turns to workability. Adding cement increases compressive strength, but decreases workability; adding water has the opposite effect.

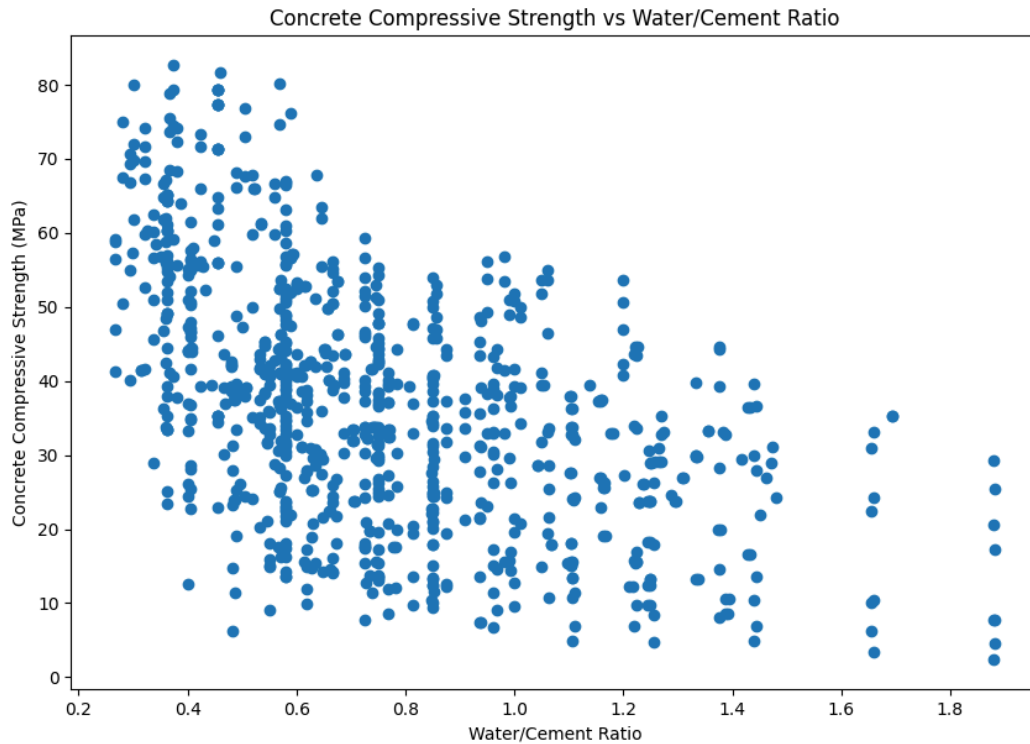


Figure 3: Strength versus water-cement ratio

The correlation between water and superplasticizer in the concrete is also visible within the data. Superplasticizer can often substitute water, increasing workability without compromising strength. Figure 4 shows the relationship of the mass of these two components in the concrete dataset. A clear negative correlation is visible from the visualization, aligning with the physical description: in practice, less water is required as superplasticizer mass increases.

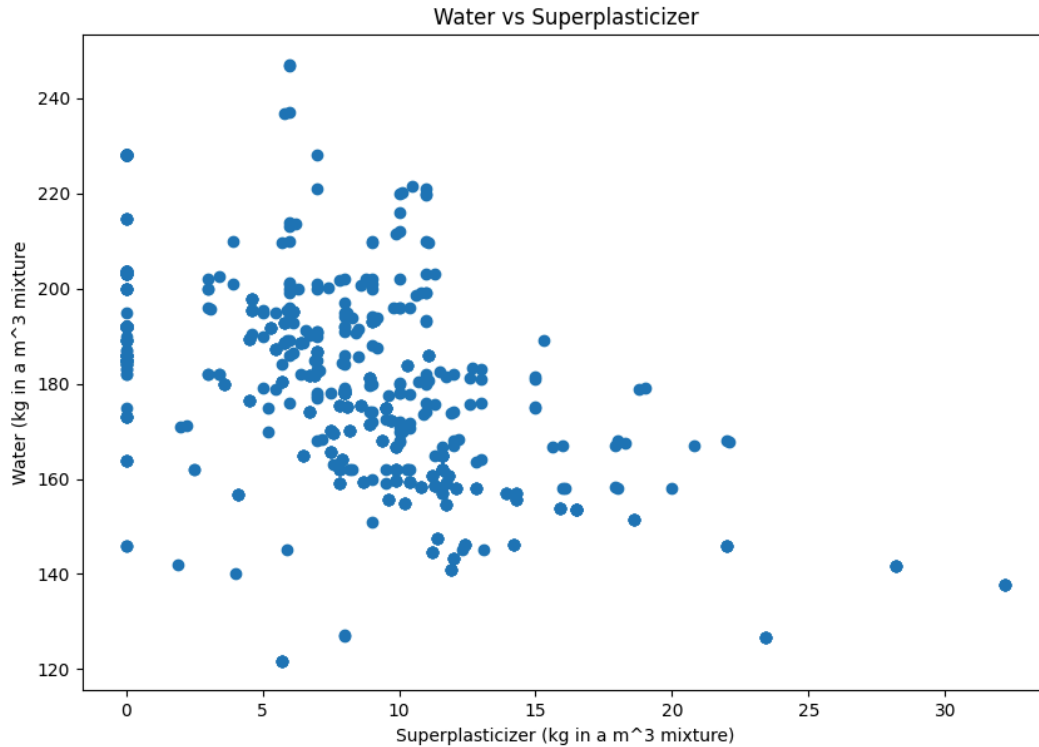


Figure 4: Water versus superplasticizer mass in concrete paste

Strategic Recommendations and Implementation Plan

Given a model has been devised that describes the compressive strength of concrete, its description can be used to create operational rules for maximizing compressive strength. The following are some suggestions from this model:

- Rule 1: To increase compressive strength, first add cement and slag in 1:1 mass ratio.
 - Justification: Both materials have the largest model coefficients, and slag typically has a lower cost per kilogram than cement.
- Rule 2: To increase workability, prefer adding plasticizer over water
 - Justification: Negative model coefficients for water mean it increases workability at the cost of compressive strength, while superplasticizer increases workability without compromising compressive strength
- Rule 3: For additional optimization to compressive strength, prefer aging to aggregates.
 - Justification: Aging has a model coefficient 3.5-4 times greater than aggregates, and the highest statistical significance of any model parameter. Furthermore, while aggregates have material costs, aging does not.

A risk mitigation plan is devised around the model given its weak predictive power. First, feature limitations are acknowledged. Namely, coarse and fine aggregates are not well defined in this model, and in practice are classified further into different grain sizes and materials. Existing

literature should be used to understand the effect that specific aggregates have on compressive strength. This model also does not account for features like temperature or humidity, which are likely to have an effect on compressive strength. The model should be assumed to be an underestimate and safety factor should be used on any predicted compressive strengths. Finally, the model predicts compressive strength, but highway projects may also require optimizing other objectives. For example, a specific porosity of concrete may be desired to absorb stormwater and avoid flooding. In such cases, further material analysis should be performed, and compressive strength should be recomputed as other objectives may lead to decreases in strength.

Given the poor predictive performance of this model, continuous monitoring of the concrete should also be performed. When concrete is being laid into the highway, some concrete should be intercepted and used to cast small samples. The compressive strength of these casts can be measured experimentally in the lab, and this figure gives an estimate to the real highway's compressive strength. Casts should be tested periodically over a month to track development, and compare against the model's predictions.